

JUBORAJ AHMED

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Education

BRAC University

Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering

2026

- **CGPA: 2.98/4.00** | **TOEFL iBT: 100** (2026)
- **Relevant coursework:** Artificial Intelligence, Neural Networks, Natural Language Processing, Data Structures and Algorithms, Operating Systems, Database Management Systems.

Research Interests

Trustworthy AI • Vision-Language Models • Federated Learning • LLM Security • AI for Healthcare • Privacy-Preserving Machine Learning

Research Experience

Trustworthy Vision-Language Models for Medical Image Understanding

BRAC University

Researcher; supervised by Dr. Ashraf Al Alam

2026–Present

- Studying reliability failures in medical vision-language models, with emphasis on sycophancy, hallucination, calibration, and robustness under misleading textual pressure.
- Developing evaluation protocols that separate harmful deference from justified response updates caused by clinical context, corrective evidence, or legitimate urgency.

JASAL: A Remote Patient Health Monitoring System Driven by Federated Learning

BRAC University

Undergraduate thesis; supervised by M. Mostakim

2025–2026

- Contributed to a three-tier federated learning framework for physiological anomaly detection using 32,796 patient records across 10 non-IID client partitions; the global model achieved 96.28% accuracy and 0.978 AUROC.
- Implemented and evaluated differential privacy and secure multiparty computation against membership-inference and gradient-inversion attacks; adaptive patient profiling reduced false-positive alarms by 49%.
- Implemented reconstruction-error-based feature attribution and INT8 quantization, obtaining 0.82 ms explanation latency and 15.6 ms end-to-end inference latency on the evaluated setup.

Research Manuscripts

- **Ahmed, J., Biswas, S. S.** “Intra-Module Subspace Coherence: A Geometric Defense Against Composite Prompt Poisoning in Federated LoRA.” *Manuscript in preparation*, 2026.

Selected Research Projects

PromptPoisonLab – LLM Security Benchmarking Toolkit

2026

- Developed a modular toolkit covering 30+ prompt-poisoning strategies, genetic-algorithm-based prompt mutation, FAISS semantic deduplication, experiment logging, and configurable risk scoring.

Prompt Injection Playground – LLM Security Evaluation Tool

2026

- Built an extensible framework for evaluating 15+ prompt-injection attacks using heuristic signals, LLM-as-a-judge scoring, and reproducible JSONL logs.

AI Context Analyst – Multi-Class Text Classification

2026

- Developed an LSTM and TF-IDF logistic-regression pipeline for multi-class text categorization, with preprocessing for 1.45M+ rows and deployment through a Dockerized Flask API.

MediTrack – AI-Powered Health Monitoring

2025

- Built a Django and PostgreSQL application integrating gradient-boosting models for predictive health-risk assessment and longitudinal visualization.

Technical Skills

Programming: Python, C++, SQL

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, NumPy, Pandas

Methods and Tools: Federated Learning, Transformers, CNNs, RNNs/LSTMs, VAEs, Model Evaluation, Git, Docker, Linux, Weights & Biases, Jupyter, LaTeX, BibTeX

Languages: Bengali (native); English (professional working proficiency)